

Online PAC Labeling from a Linear Programming Perspective

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1 Problem Setup and LP Formulation

There is a fixed pool of unlabeled items $X_1, \dots, X_T$, whose true labels $Y_1, \dots, Y_T$ are initially unknown, and the goal is to labeling the items correctly while minimizing labeling cost.

Define the action set $\mathcal{A} = \{E, 1, \dots, k\}$, where E denotes deferral to the expert. The expert label Y_t is treated as correct but incurs an expansive query cost $c_t(E) > 0$. For each item X_t , we have access to k cheap ML models, which produce predicted labels $\hat{Y}_t^{(j)}$.

In this project, we assume ML models are much cheaper than the expert. Therefore, define the loss $h_t(a)$ and cost $c_t(a)$ for action a on item X_t by

$$h_t(a) = \begin{cases} 0, & a = E, \\ \ell(Y_t, \hat{Y}_t^{(a)}), & a \in [k], \end{cases} \quad c_t(a) = \begin{cases} 1, & a = E, \\ 0, & a \in [k]. \end{cases} \quad (1)$$

The labeling loss is assumed bounded and normalized: $0 \leq \ell(Y_t, \hat{Y}_t^{(j)}) \leq 1, \forall t \in [T], j \in [k]$.

A labeling rule chooses an action $a_t \in \mathcal{A}$ for each item. The resulting final label is

$$\tilde{Y}_t = \begin{cases} Y_t, & a_t = E, \\ \hat{Y}_t^{(a_t)}, & a_t \in [k]. \end{cases}$$

1.1 The Cost-Aware Labeling LP

We formulate the cost-aware PAC labeling problem as a linear program. Intuitively, the LP minimizes labeling cost subject to an average-error constraint. The LP explicitly takes both the cost and the loss into account, and it models multi-source setting.

Let $x_{t,a}$ be the probability of choosing action a on item t . If the true losses $h_t(a)$ were known, the optimal labeling policy would be the solution to the following LP:

$$\begin{aligned} \text{minimize}_x \quad & \frac{1}{T} \sum_{t=1}^T \sum_{a \in \mathcal{A}} c_t(a) x_{t,a} \quad \text{subject to} \quad \frac{1}{T} \sum_{t=1}^T \sum_{a \in \mathcal{A}} h_t(a) x_{t,a} \leq \varepsilon, \\ & \sum_{a \in \mathcal{A}} x_{t,a} = 1, \quad t \in [T], \\ & x_{t,a} \geq 0, \quad t \in [T], a \in \mathcal{A}. \end{aligned} \quad (2)$$

1.2 Lagrangian Dual and Shadow Price

The oracle LP (2) has a single global average-error constraint. Let $\lambda \geq 0$ denote the Lagrange multiplier, or shadow price, of the labeling error constraint. The dual problem is the one-dimensional concave maximization problem (See Appendix B.1 for details):

$$\max_{\lambda \geq 0} D(\lambda) := -\lambda \varepsilon + \frac{1}{T} \sum_{t=1}^T \min_{a \in \mathcal{A}} \{c_t(a) + \lambda h_t(a)\}. \quad (3)$$

This is the source of the fixed-price rule: once a price λ is fixed, the dual minimization separates across items, and the action for item t is given by the following simple rule:

$$a_t(\lambda) \in \arg \min_{a \in \mathcal{A}} \{c_t(a) + \lambda h_t(a)\}. \quad (4)$$

The dual price λ balances the trade-off between cost and error.

Roadmap The LP view reduces the expert-error trade-off to a single dual price λ . Because the oracle rule depends on the unknown losses $h_t(a)$, we use proxy losses $\hat{h}_t(a)$, learned on an independent split or fixed in advance. These proxies need not be conservative or calibrated: better proxies improve cost, while held-out calibration provides validity. We assume the itemwise losses in the argmin rules are distinct, so ties do not occur.

Section 2 treats the **offline** case: estimate a certified price on a calibration set, then label the target pool at $\hat{\lambda}$. Section 3 treats the **online** case, where items arrive sequentially, audits reveal labels intermittently, and the price λ_t is updated from audited feedback.

2 Offline PAC Labeling

We first use a labeled calibration sample to select a certified price $\hat{\lambda}$, then label the target pool at this fixed price.

2.1 Price Estimation on a Calibration Set

For price estimation, we form the plug-in policy family by replacing the unknown loss $h_t(a)$ in (4) with the fixed proxy loss $\hat{h}_t(a)$. In the equal-cost setting $c_t(j) = 0$ and $c_t(E) = 1$, the chosen cheap source is independent of λ : first set

$$j_t^\lambda = j_t^* \in \arg \min_{j \in [k]} \hat{h}_t(j), \quad s_t = \hat{h}_t(j_t^*) = \min_{j \in [k]} \hat{h}_t(j). \quad (5)$$

The price λ then only decides whether to accept this routed cheap label or defer to the expert:

$$D_t^\lambda = \mathbf{1}\{s_t \geq 1/\lambda\}, \quad a_t^\lambda = \begin{cases} E, & D_t^\lambda = 1, \\ j_t^*, & D_t^\lambda = 0, \end{cases} \quad (6)$$

with the conventions $1/0 = \infty$ and $1/\infty = 0$. Equivalently,

$$a_t^\lambda \in \arg \min_{a \in \mathcal{A}} \{c_t(a) + \lambda \hat{h}_t(a)\}.$$

The realized labeling loss and the cost given a price λ are then

$$H_t(\lambda) = (1 - D_t^\lambda) \ell(Y_t, \hat{Y}_t^{(j_t^*)}), \quad C_t(\lambda) = D_t^\lambda. \quad (7)$$

Then the average loss and cost of the price- λ policy are

$$L_T(\lambda) = \frac{1}{T} \sum_{t=1}^T H_t(\lambda), \quad C_T(\lambda) = \frac{1}{T} \sum_{t=1}^T C_t(\lambda). \quad (8)$$

Here $C_T(\lambda)$ is known from the proxies and costs, while $L_T(\lambda)$ depends on true labels. The calibration set is used exactly to estimate this loss.

We sample calibration indices $I_1, \dots, I_m$ independently and uniformly from the target pool, with replacement, and collect the corresponding expert labels. Define the empirical labeling loss

$$\hat{L}_m(\lambda) = \frac{1}{m} \sum_{s=1}^m H_{I_s}(\lambda). \quad (9)$$

For any fixed λ , Hoeffding's inequality gives a one-sided upper confidence bound on $L_T(\lambda)$:

$$U_m(\lambda; \alpha) = \hat{L}_m(\lambda) + \sqrt{\frac{\log(1/\alpha)}{2m}}. \quad (10)$$

We do not need a union bound over λ here: as λ increases, the threshold $1/\lambda$ decreases, so each item can only move from the routed AI model to the expert and cannot switch between cheap models. Thus $H_t(\lambda)$ is non-increasing and $C_t(\lambda)$ is non-decreasing in λ . The certified price is

$$\hat{\lambda} = \min \{ \lambda \geq 0 : U_m(\lambda; \alpha) \leq \varepsilon \}, \quad (11)$$

falling back to the expert-only policy $\lambda = \infty$ if no finite price qualifies. Because $C_T(\lambda)$ is non-decreasing and $U_m(\lambda; \alpha)$ non-increasing in λ , (11) coincides with the cost-minimizing selection $\arg \min_{\lambda \in [0, \infty]} \{ C_T(\lambda) : U_m(\lambda; \alpha) \leq \varepsilon \}$.

2.2 Labeling the Pool at Price $\hat{\lambda}$

After calibration, the price is fixed. The final labeling rule is the plug-in dual rule at $\hat{\lambda}$:

$$a_t^{\hat{\lambda}} \in \arg \min_{a \in \mathcal{A}} \{ c_t(a) + \hat{\lambda} \hat{h}_t(a) \}.$$

In the equal-cost form this rule is especially simple. For each target item, compute j_t^* and s_t from (5). If $s_t \geq 1/\hat{\lambda}$, query the expert and release $\tilde{Y}_t = Y_t$. Otherwise, accept the routed cheap label and release $\tilde{Y}_t = \hat{Y}_t^{(j_t^*)}$.

Relation to PAC labeling. In the single-source case, using uncertainty score as the proxy $\hat{h}_t = U_t$, the AI label is used when $\hat{h}_t = U_t < 1/\hat{\lambda}$. Thus the LP rule recovers the original PAC-labeling threshold family [1]; with multiple sources, U_t is replaced by $\min_j \hat{h}_t(j)$.

2.3 Offline PAC Guarantee

Theorem 1 (Offline monotone-threshold PAC guarantee). *Let $\hat{\lambda}$ be selected over $[0, \infty]$ by (11). Then, with probability at least $1 - \alpha$ over the draw of the calibration indices, the released labels satisfy*

$$\frac{1}{T} \sum_{t=1}^T \ell(Y_t, \tilde{Y}_t) \leq \varepsilon. \quad (12)$$

The proof is in Appendix B.2.

3 Online PAC Labeling via Audits

In the online setting, inputs arrive sequentially and AI losses are unobserved unless an expert label is revealed. We therefore keep a moving PAC-labeling threshold and use randomized audits for intermittent feedback.

3.1 Online PAC Labeling Setup and Audited Feedback

Following the action-cost notation in (1), we focus on the same regime where $c_t(j) = 0 \forall t, j \in [k]$, $c_t(E) = 1$. The goal is to label as many items as possible with AI while controlling the average error of the released dataset.

At time t , the learner observes the input X_t , cheap model labels $\{\hat{Y}_t^{(j)}\}_{j=1}^k$, and predictable proxy losses $\{\hat{h}_t(j)\}_{j=1}^k$. The proxy losses may be uncertainty scores, calibrated error estimates, or any predictable scores built from past expert/audit labels and current non-expert information. The algorithm first routes the item to the model with the smallest proxy loss:

$$j_t^\lambda \in \arg \min_{j \in [k]} \hat{h}_t(j), \quad \hat{h}_t^\lambda = \hat{h}_t(j_t^\lambda). \quad (13)$$

It then applies a scalar PAC-labeling threshold. Let $\lambda_t \geq 0$ denote the current price of error. Equivalently, $1/\lambda_t$ is the current uncertainty threshold, with the convention that $1/\lambda_t = \infty$ when $\lambda_t = 0$. The deferral decision is

$$D_t = \mathbf{1}\{\lambda_t \hat{h}_t^\lambda \geq 1\} = \mathbf{1}\{\hat{h}_t^\lambda \geq 1/\lambda_t\}. \quad (14)$$

If $D_t = 1$, the item is sent to the expert and the released label is $\tilde{Y}_t = Y_t$. Otherwise it initially releases the routed AI label. This is the PAC-labeling threshold form, with the multi-model extension contained in $\hat{h}_t^\lambda = \min_j \hat{h}_t(j)$.

When an AI label is used, its true loss is unobserved unless the expert label is also revealed. After every AI decision, the algorithm uses an audit probability chosen before the audit draw and before observing the current expert label:

$$A_t \sim \text{Bernoulli}(p_t), \quad p_t \in [p_{\min}, 1].$$

If $A_t = 1$, the expert label is queried, the final released label is corrected to Y_t , and the audit records the counterfactual error of the routed AI label.

Deferral has zero loss because the expert label is used. The update uses an internal target $\varepsilon_0 \leq \varepsilon$:

$$\hat{Z}_t = \begin{cases} -\varepsilon_0, & \text{if the item is deferred to the expert,} \\ \{\ell(Y_t, \hat{Y}_t^{(j_t^\lambda)}) - \varepsilon_0\}/p_t, & \text{if the AI label is used and audited,} \\ 0, & \text{if the AI label is used and not audited.} \end{cases}$$

The role of ε_0 is to leave room for the finite-horizon guarantee: the algorithm updates toward the internal target ε_0 , while the final PAC target is ε . Corollary 2 gives the condition under which choosing ε_0 yields final error at most ε . The price update is

$$\lambda_{t+1} = [\lambda_t + \eta \hat{Z}_t]_+, \quad (15)$$

where $\eta > 0$ is a step size. Audited mistakes increase λ_t , lowering $1/\lambda_t$ and making future routing more conservative; deferrals or low-error audits decrease it.

Let $G_t = \ell(Y_t, \tilde{Y}_t)$ be the realized final loss after audit-and-correct. Since audited AI errors and expert deferrals use the true label,

$$G_t \leq H_t.$$

Thus G_t is the final audit-and-correct loss, while H_t tracks the counterfactual AI loss used for feedback.

Algorithm 1 Audited online PAC labeling

Require: Stream horizon T ; target error $\varepsilon \in [0, 1]$; audit floor $p_{\min}$; step size η ; internal target $\varepsilon_0 \in [0, \varepsilon]$.

```
1: Initialize  $\lambda_1 \geq 0$ .
2: for  $t = 1, \dots, T$  do
3:   Observe  $X_t$ , cheap labels  $\{\hat{Y}_t^{(j)}\}_{j=1}^k$ , and predictable proxies  $\{\hat{h}_t(j)\}_{j=1}^k$ .
4:   Choose  $j_t^\lambda$  and  $\hat{h}_t^\lambda$  by (13).
5:   if  $\lambda_t \hat{h}_t^\lambda \geq 1$  then
6:     Query the expert, output  $\tilde{Y}_t = Y_t$ , set  $D_t = 1$  and  $\hat{Z}_t = -\varepsilon_0$ .
7:   else
8:     Output the AI label  $\tilde{Y}_t = \hat{Y}_t^{(j_t^\lambda)}$  and set  $D_t = 0$ .
9:     Choose  $p_t \in [p_{\min}, 1]$  before observing  $Y_t$ , then draw  $A_t \sim \text{Bernoulli}(p_t)$ .
10:    if  $A_t = 1$  then
11:      Query the expert, correct  $\tilde{Y}_t = Y_t$ , and set  $\hat{Z}_t = \{\ell(Y_t, \hat{Y}_t^{(j_t^\lambda)}) - \varepsilon_0\}/p_t$ .
12:    else
13:      Set  $\hat{Z}_t = 0$ .
14:    end if
15:  end if
16:  Update  $\lambda_{t+1}$  by (15).
17: end for
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3.2 Online Finite-Horizon Guarantee

The online result is distribution-free with respect to the data sequence; the probability is over the algorithm's audit randomness. It controls realized error, not regret against an offline expert-query oracle.

Assumption 1 (Predictable routing and random audits). *The routing decision, audit probability, and selected source are chosen using only past information and the current non-expert information $(X_t, \{\hat{Y}_t^{(j)}, \hat{h}_t(j)\}_{j=1}^k)$. After this decision, the audit draw is independent Bernoulli with known probability $p_t \in [p_{\min}, 1]$.*

Assumption 2 (Proxy floor). *For some $\rho > 0$, $\hat{h}_t(j) \in [\rho, 1]$ for all t, j .*

For the finite-horizon analysis, define the augmented observation mask and its conditional observation probability by

$$O_t = D_t + (1 - D_t)A_t, \quad q_t = D_t + (1 - D_t)p_t. \quad (16)$$

Thus $q_t = 1$ on expert deferrals and $q_t = p_t$ on AI decisions. The three-case update in Algorithm 1 can equivalently be written as the centered inverse-observation signal

$$\hat{Z}_t = \frac{O_t}{q_t} (H_t - \varepsilon_0). \quad (17)$$

Let $\mathcal{G}_t$ denote the pre-audit sigma-field that contains the history through round $t - 1$, the current non-expert information, the selected source j_t^λ , the deferral decision D_t , the audit probability p_t , and the latent selected loss. It does not contain the current audit draw A_t . The audit assumption says that, conditional on $\mathcal{G}_t$, A_t is an independent Bernoulli random variable with mean p_t whenever $D_t = 0$.

Proposition 1 (Unbiased intermittent update signal). *Under bounded loss and online predictability,*

$$\mathbb{E}\left[\frac{O_t}{q_t} \middle| \mathcal{G}_t\right] = 1, \quad \mathbb{E}[\widehat{Z}_t \mid \mathcal{G}_t] = H_t - \varepsilon_0.$$

Theorem 2 (Audited online finite-horizon control). *Assume bounded loss, online predictability and audits, and the proxy floor. Let*

$$\lambda_{\text{def}} = \frac{1}{\rho}, \quad \Lambda_{\text{on}} = \max\left\{\lambda_1, \lambda_{\text{def}} + \frac{\eta(1 - \varepsilon_0)}{p_{\min}}\right\}.$$

Run Algorithm 1 with constant step size $\eta > 0$ and internal target $\varepsilon_0 \in [0, \varepsilon]$, where $\varepsilon \in [0, 1]$. Then, for any fixed horizon T and any $\delta \in (0, 1)$, with probability at least $1 - \delta$ over the audit randomness,

$$\frac{1}{T} \sum_{t=1}^T G_t \leq \frac{1}{T} \sum_{t=1}^T H_t \leq \varepsilon_0 + \frac{\Lambda_{\text{on}}}{\eta T} + \frac{1}{p_{\min}} \sqrt{\frac{2 \log(1/\delta)}{T}}. \quad (18)$$

where G_t is the final audit-and-correct loss, while H_t tracks the counterfactual AI loss used for feedback.

The proofs of Proposition 1, Theorem 2, and Corollary 2 are in Appendix B.3.

4 Experiments

We run a proof-of-concept study on ImageNet and ImageNetV2 using the provided confidence score, with the model prediction as the cheap source and the ground-truth label as the expert. Table 1 reports representative results at $\varepsilon = 0.10$; all reported numbers average over 500 random trials. The experiments check three basic points.

1. **One source.** With a single cheap source, the LP fixed-price rule reduces to the same scalar threshold family as the original PAC labeling method, and empirically recovers the same behavior.
2. **Two sources.** We then add a second cheap source to create a simple routing problem. The LP rule can choose the lower-proxy source on each item before deciding whether to defer to the expert, improving budget savings over the best calibrated single-source baseline.
3. **Online.** Finally, the audited online update also shows the desired validity–cost tradeoff under partial feedback.

Full sweeps and implementation details are deferred to Appendix C.

References

- [1] Emmanuel J. Candès, Andrew Ilyas, and Tijana Zrnic. Probably approximately correct labels. *arXiv preprint arXiv:2506.10908*, 2025.
- [2] Vladimir Vovk, Alexander Gammernan, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.
- [3] Anastasios N. Angelopoulos and Stephen Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *arXiv preprint arXiv:2107.07511*, 2021.

Setting	Dataset	Method	Error	Violation	Budget saved
One source	ImageNet	LP fixed price	0.0873	0.0000	0.7848
One source	ImageNet	Original PAC	0.0879	0.0000	0.7864
One source	ImageNetV2	LP fixed price	0.0718	0.0000	0.5457
One source	ImageNetV2	Original PAC	0.0726	0.0000	0.5479
Two sources	ImageNet	LP routing	0.0873	0.0000	0.7848
Two sources	ImageNet	Best single source	0.0879	0.0000	0.4746
Two sources	ImageNetV2	LP routing	0.0718	0.0000	0.5457
Two sources	ImageNetV2	Best single source	0.0720	0.0000	0.3587
Online	ImageNet	Audited, $p = 0.2$	0.0737	0.0000	0.6115
Online	ImageNetV2	Audited, $p = 0.2$	0.0783	0.0000	0.4681

Table 1: Representative averages at $\varepsilon = 0.10$.

- [4] Isaac Gibbs and Emmanuel Candès. Adaptive conformal inference under distribution shift. In *Advances in Neural Information Processing Systems*, 2021.
- [5] Haosen Ge, Hamsa Bastani, and Osbert Bastani. Stochastic online conformal prediction with semi-bandit feedback. *arXiv preprint arXiv:2405.13268*, 2024.
- [6] Bowen Wang, Matteo Zecchin, and Osvaldo Simeone. Mirror online conformal prediction with intermittent feedback. *arXiv preprint arXiv:2503.10345*, 2025.
- [7] Junyoung Yang, Kyungmin Kim, and Sangdon Park. Online conformal prediction with adversarial semi-bandit feedback via regret minimization. *arXiv preprint arXiv:2604.17984*, 2026.
- [8] David Madras, Toniann Pitassi, and Richard Zemel. Predict responsibly: Improving fairness and accuracy by learning to defer. In *Advances in Neural Information Processing Systems*, 2018.
- [9] Hussein Mozannar and David Sontag. Consistent estimators for learning to defer to an expert. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 7076–7087, 2020.
- [10] Anqi Mao, Christopher Mohri, Mehryar Mohri, and Yutao Zhong. Two-stage learning to defer with multiple experts. In *Advances in Neural Information Processing Systems*, 2023.
- [11] Jean V. Alves, Diogo Leitão, Sérgio Jesus, Marco O. P. Sampaio, Javier Liébana, Pedro Saleiro, Mário A. T. Figueiredo, and Pedro Bizarro. Cost-sensitive learning to defer to multiple experts with workload constraints. *Transactions on Machine Learning Research*, 2024.

A Related Work

PAC labels and conformal calibration. The closest starting point is PAC labeling [1], which gives finite-sample guarantees for cost-effective dataset labeling using AI predictions and expert audits. That framework already includes both scalar-threshold PAC labeling and multi-model PAC routing variants. The calibration logic is closely related to conformal prediction and distribution-free uncertainty quantification [2, 3], but the object being controlled here is the average error of a released labeled dataset rather than coverage of prediction sets.

Online conformal prediction with partial feedback. Adaptive and online conformal methods update calibration parameters over time to handle changing environments [4]. Recent work has studied intermittent or semi-bandit feedback, where labels are not always observed [5, 6, 7]. Those papers motivate the audited online extension in Section 3.

Learning to defer and cost-sensitive routing. Learning-to-defer methods train systems that either predict or pass an item to an expert [8, 9]. Multiple-expert and cost-sensitive variants study richer deferral and routing settings [10, 11]. The distinction here is the finite-sample PAC certificate for the final released labels. The proxy router may be learned by standard supervised methods, but the guarantee comes from held-out calibration of the actual selected policy.

B Proofs

B.1 Dual Derivation

Introduce one dual variable $\lambda \geq 0$ for the average-error constraint and one free dual variable $\mu_t \in \mathbb{R}$ for each simplex equality. The nonnegativity constraints $x_{t,a} \geq 0$ are kept as the domain of the inner minimization.

Primal constraint	Dual variable	Sign restriction
$\frac{1}{T} \sum_{t,a} h_t(a) x_{t,a} \leq \varepsilon$	λ	$\lambda \geq 0$
$\sum_a x_{t,a} = 1$ for item t	μ_t	$\mu_t \in \mathbb{R}$
$x_{t,a} \geq 0$	domain constraint	–

Using the equality convention $\mu_t(1 - \sum_a x_{t,a})$, the Lagrangian is

$$\begin{aligned} \mathcal{L}(x; \lambda, \mu) &= \frac{1}{T} \sum_{t,a} c_t(a) x_{t,a} + \lambda \left(\frac{1}{T} \sum_{t,a} h_t(a) x_{t,a} - \varepsilon \right) + \sum_{t=1}^T \mu_t \left(1 - \sum_a x_{t,a} \right) \\ &= \sum_{t,a} \left[\frac{1}{T} \{c_t(a) + \lambda h_t(a)\} - \mu_t \right] x_{t,a} + \sum_{t=1}^T \mu_t - \lambda \varepsilon. \end{aligned} \quad (19)$$

For fixed (λ, μ) , the inner problem is $\inf_{x \geq 0} \mathcal{L}(x; \lambda, \mu)$. Since the Lagrangian is linear in every $x_{t,a}$, this infimum is finite if and only if

$$\mu_t \leq \frac{1}{T} \{c_t(a) + \lambda h_t(a)\}, \quad t \in [T], \quad a \in \mathcal{A}. \quad (20)$$

Otherwise some variable with a negative coefficient can be sent to $+\infty$, driving the Lagrangian to $-\infty$. When (20) holds, the x terms vanish at the inner minimum and the dual function is

$$g(\lambda, \mu) = \sum_{t=1}^T \mu_t - \lambda \varepsilon.$$

Thus the explicit dual LP is

$$\begin{aligned} &\text{maximize}_{\lambda, \mu} \quad \sum_{t=1}^T \mu_t - \lambda \varepsilon \\ &\text{subject to} \quad \mu_t \leq \frac{1}{T} \{c_t(a) + \lambda h_t(a)\}, \quad t \in [T], \quad a \in \mathcal{A}, \\ &\quad \lambda \geq 0, \quad \mu_t \in \mathbb{R}, \quad t \in [T]. \end{aligned} \quad (21)$$

For any fixed λ , the objective is increasing in each μ_t , so the optimal value is attained at

$$\mu_t^*(\lambda) = \frac{1}{T} \min_{a \in \mathcal{A}} \{c_t(a) + \lambda h_t(a)\}.$$

Substitution gives the reduced dual (3). The primal is feasible because choosing the expert for all items gives zero loss, and it is bounded because costs are nonnegative and finite. Therefore strong duality holds by finite-dimensional linear programming duality.

B.2 Offline Proofs and Extended Results

Proof of Theorem 1. Condition on the fixed target pool, labels, and proxy losses from the theorem. The routed source $j_t^* \in \arg \min_j \hat{h}_t(j)$, the routed score $s_t = \hat{h}_t(j_t^*)$, and the routed loss

$$r_t = \ell(Y_t, \hat{Y}_t^{(j_t^*)}) \in [0, 1]$$

are fixed. Under equal cheap-source costs the routed source does not depend on λ ; the price enters only through

$$D_t^\lambda = \mathbf{1}\{s_t \geq 1/\lambda\}, \quad H_t(\lambda) = r_t \mathbf{1}\{s_t < 1/\lambda\}.$$

Thus as λ increases, each item can only move from routed to deferred, so $H_t(\lambda)$, $L_T(\lambda)$, and $\hat{L}_m(\lambda)$ are non-increasing in λ .

For each fixed λ , the calibration draws $H_{I_s}(\lambda)$ are i.i.d. in $[0, 1]$ with mean $L_T(\lambda)$. Hoeffding's inequality gives

$$\mathbb{P}\{L_T(\lambda) > U_m(\lambda; \alpha)\} \leq \alpha, \quad U_m(\lambda; \alpha) = \hat{L}_m(\lambda) + \sqrt{\frac{\log(1/\alpha)}{2m}}.$$

This is a pointwise Hoeffding statement; the next monotonicity step is what removes the need for a union bound over prices.

The equal-cost policies form a nested finite family on the fixed target pool: as λ varies over $[0, \infty]$, the routed set $\{t : s_t < 1/\lambda\}$ changes only when the threshold $1/\lambda$ crosses one of the finitely many scores s_t . Order the distinct induced policies by increasing λ . If every induced policy has $L_T(\lambda) \leq \varepsilon$, the claim is immediate. Otherwise, let λ° be any price inducing the last policy in this order whose true loss is still above ε :

$$L_T(\lambda^\circ) > \varepsilon, \quad L_T(\lambda) \leq \varepsilon \quad \text{for every later policy.}$$

This λ° is fixed conditional on the target pool and labels, not chosen from the calibration sample. If the selected price were bad, i.e. $L_T(\hat{\lambda}) > \varepsilon$, then $\hat{\lambda}$ would induce a policy no later than λ° . Since $U_m(\lambda; \alpha)$ is non-increasing in λ ,

$$U_m(\lambda^\circ; \alpha) \leq U_m(\hat{\lambda}; \alpha) \leq \varepsilon < L_T(\lambda^\circ).$$

Thus the bad-selection event is contained in the single fixed-price failure event $\{L_T(\lambda^\circ) > U_m(\lambda^\circ; \alpha)\}$, which has probability at most α . Therefore $\mathbb{P}\{L_T(\hat{\lambda}) \leq \varepsilon\} \geq 1 - \alpha$.

Finally, the released label equals $\hat{Y}_t^{(j_t^*)}$ on routed items and Y_t on deferred items, so $\frac{1}{T} \sum_t \ell(Y_t, \tilde{Y}_t) = L_T(\hat{\lambda})$. If calibration items are corrected to their expert labels, the realized loss only decreases. $\square$

The following corollary gives a population risk control under a same-distribution assumption.

Corollary 1 (Generalization guarantee). *Suppose the calibration set is an i.i.d. sample from a distribution P . For a price λ , define the population risk*

$$L(\lambda) = \mathbb{E}_{(X,Y) \sim P} \left[(1 - D^\lambda(X)) \ell(Y, \hat{Y}^{(j^*(X))}(X)) \right].$$

Assume the equal-cost monotone-threshold policy class above, and let $\hat{\lambda}$ be selected over $[0, \infty]$ by (11). Then $L(\hat{\lambda}) \leq \varepsilon$ with probability at least $1 - \alpha$. Moreover, for an independent deployment pool of size n , with probability at least $1 - \alpha - \delta$,

$$\frac{1}{n} \sum_{t=1}^n \ell(Y_t, \tilde{Y}_t) \leq \varepsilon + \sqrt{\frac{\log(1/\delta)}{2n}}. \quad (22)$$

Proof of Corollary 1. Under equal cheap-source costs the routed source $j^*(X)$ does not depend on λ , and

$$H_\lambda(X, Y) = (1 - D^\lambda(X)) \ell(Y, \hat{Y}^{(j^*(X))}(X))$$

is pointwise non-increasing in λ . Hence $L(\lambda) = \mathbb{E} H_\lambda$ and $\hat{L}_m(\lambda)$ are non-increasing. For each fixed λ , the calibration losses are i.i.d. in $[0, 1]$ with mean $L(\lambda)$, so Hoeffding's fixed-price bound applies.

Let $\lambda_\star = \sup\{\lambda : L(\lambda) > \varepsilon\}$, with the claim immediate if this set is empty. If $L(\lambda_\star) > \varepsilon$, the same single-point argument as in Theorem 1 applies at $\lambda_\star$. If $L(\lambda_\star) \leq \varepsilon$, choose a deterministic increasing sequence $\lambda_n \uparrow \lambda_\star$ with $L(\lambda_n) > \varepsilon$. On the bad event $L(\hat{\lambda}) > \varepsilon$, some λ_n satisfies $\lambda_n \geq \hat{\lambda}$, and therefore

$$U_m(\lambda_n; \alpha) \leq U_m(\hat{\lambda}; \alpha) \leq \varepsilon < L(\lambda_n).$$

The events $\{U_m(\lambda_n; \alpha) \leq \varepsilon\}$ are increasing in n , and for each fixed n Hoeffding gives

$$\mathbb{P}\{U_m(\lambda_n; \alpha) \leq \varepsilon\} \leq \mathbb{P}\{U_m(\lambda_n; \alpha) < L(\lambda_n)\} \leq \alpha.$$

Taking the limit over n gives $L(\hat{\lambda}) \leq \varepsilon$ with probability at least $1 - \alpha$.

Condition on the calibration sample and on the event $L(\hat{\lambda}) \leq \varepsilon$. The deployment losses $\ell(Y_t, \tilde{Y}_t)$ are then independent, bounded in $[0, 1]$, and have conditional mean at most $L(\hat{\lambda}) \leq \varepsilon$, so Hoeffding's inequality gives

$$\frac{1}{n} \sum_{t=1}^n \ell(Y_t, \tilde{Y}_t) \leq \varepsilon + \sqrt{\frac{\log(1/\delta)}{2n}}$$

with conditional probability at least $1 - \delta$. Combining the calibration and deployment events gives probability at least $1 - \alpha - \delta$. $\square$

B.3 Online Proofs

Proof of Proposition 1. Recall that $\mathcal{G}_t$ fixes the predictable round- t information, the selected source j_t^λ , the routed proxy loss $\hat{h}_t^\lambda = \hat{h}_t(j_t^\lambda)$, the audit probability p_t , the deferral decision D_t , and the latent selected loss, but not the audit draw A_t . If $D_t = 1$, then $O_t = q_t = 1$ and $H_t = 0$, so

$$\mathbb{E} \left[\frac{O_t}{q_t} \middle| \mathcal{G}_t \right] = 1, \quad \mathbb{E}[\hat{Z}_t \mid \mathcal{G}_t] = -\varepsilon_0 = H_t - \varepsilon_0.$$

If $D_t = 0$, then $O_t = A_t$, $q_t = p_t$, and conditional on $\mathcal{G}_t$ the audit draw is Bernoulli(p_t) while the loss $\ell(Y_t, \hat{Y}_t^{(j_t^\lambda)})$ is fixed. Hence

$$\mathbb{E} \left[\frac{O_t}{q_t} \middle| \mathcal{G}_t \right] = \frac{p_t}{p_t} = 1.$$

Since H_t and ε_0 are fixed conditional on $\mathcal{G}_t$, this also gives

$$\mathbb{E}[\widehat{Z}_t \mid \mathcal{G}_t] = \mathbb{E}\left[\frac{O_t}{q_t}(H_t - \varepsilon_0) \mid \mathcal{G}_t\right] = H_t - \varepsilon_0.$$

□

Proof of Theorem 2. First bound the price process. If $\lambda_t \geq \lambda_{\text{def}} = 1/\rho$, then

$$\lambda_t \widehat{h}_t^\lambda = \lambda_t \widehat{h}_t(j_t^\lambda) \geq \lambda_{\text{def}} \rho = 1.$$

Thus the algorithm defers to the expert, so $D_t = 1$ and $\widehat{Z}_t = -\varepsilon_0$. Consequently $\lambda_{t+1} = [\lambda_t - \eta \varepsilon_0]_+ \leq \lambda_t$. If instead $\lambda_t < \lambda_{\text{def}}$, then $\widehat{Z}_t \leq (1 - \varepsilon_0)/p_{\min}$ because $\varepsilon_0 \in [0, 1]$ and the loss is bounded by one. Hence

$$\lambda_{t+1} \leq \lambda_t + \eta(1 - \varepsilon_0)/p_{\min} < \lambda_{\text{def}} + \eta(1 - \varepsilon_0)/p_{\min}.$$

By induction, $0 \leq \lambda_t \leq \Lambda_{\text{on}}$ for all t .

Since $[x]_+ \geq x$, the update gives

$$\lambda_{t+1} \geq \lambda_t + \eta \widehat{Z}_t.$$

Summing over $t = 1, \dots, T$ yields

$$\sum_{t=1}^T \widehat{Z}_t \leq \frac{\lambda_{T+1} - \lambda_1}{\eta} \leq \frac{\Lambda_{\text{on}}}{\eta}.$$

Let $M_t = (H_t - \varepsilon_0) - \widehat{Z}_t$. By Proposition 1 and iterated expectation, M_t is a martingale difference with respect to the filtration generated by the past rounds and the current audit draw. Moreover, $|M_t| \leq 1/p_{\min}$: if $D_t = 1$, then $M_t = 0$; if $D_t = 0$ and $A_t = 0$, then $|M_t| = |H_t - \varepsilon_0| \leq 1$; and if $D_t = 0$ and $A_t = 1$, then

$$|M_t| = |H_t - \varepsilon_0| \left| \frac{1}{p_t} - 1 \right| \leq \frac{1}{p_t} \leq \frac{1}{p_{\min}}.$$

Azuma-Hoeffding gives, with probability at least $1 - \delta$,

$$\sum_{t=1}^T M_t \leq \frac{1}{p_{\min}} \sqrt{2T \log(1/\delta)}.$$

Because

$$\sum_{t=1}^T H_t = T\varepsilon_0 + \sum_{t=1}^T \widehat{Z}_t + \sum_{t=1}^T M_t,$$

combining the last displays gives (18). The same display includes the final-loss bound because audit-and-correct gives $G_t \leq H_t$ pointwise. □

Corollary 2 (PAC-style online tuning). *Under the conditions of Theorem 2, let $\bar{\Lambda}_{\text{on}} = \max\{\lambda_1, 1/\rho + \eta/p_{\min}\}$ and*

$$B_T = \frac{\bar{\Lambda}_{\text{on}}}{\eta T} + \frac{1}{p_{\min}} \sqrt{\frac{2 \log(1/\delta)}{T}}.$$

If $B_T \leq \varepsilon$, choose the internal target $\varepsilon_0 \in [0, \varepsilon - B_T]$, typically $\varepsilon_0 = \varepsilon - B_T$. Then the audit-and-correct final labels satisfy $\frac{1}{T} \sum_{t=1}^T G_t \leq \varepsilon$ with probability at least $1 - \delta$. With $\eta \asymp T^{-1/2}$ and constant $p_{\min}$, $B_T = O(T^{-1/2})$.

Proof of Corollary 2. Since $\varepsilon_0 \geq 0$, we have $\Lambda_{\text{on}} \leq \bar{\Lambda}_{\text{on}}$. Hence the choice $\varepsilon_0 \leq \varepsilon - B_T$ makes the right-hand side of (18) at most ε . If $\eta \asymp T^{-1/2}$, $p_{\min}$ is constant, and the proxy floor is fixed, then $B_T = O(T^{-1/2})$. $\square$

C Experimental Details

The experiments use two ImageNet prediction tables: ImageNet has $T = 50000$ items and ImageNetV2 has $T = 10000$ items. Each file contains a ground-truth label, a model prediction, and a confidence score. We treat the model prediction as the only cheap source and the ground-truth label as the expert. The proxy loss is $\hat{h} = 1 - \text{confidence}$, the cheap-source deployment cost is zero, and the expert cost is one. All experiments use 500 random trials.

Offline implementation. For the fixed-price method, the candidate threshold grid contains the expert-only rule plus 201 evenly spaced confidence-loss thresholds between 0 and 1. This grid is an implementation choice for the sweep; the monotone certificate in Section 2.1 does not require prices to be restricted to a finite grid. The experiment uses the monotone threshold radius

$$\sqrt{\frac{\log(1/\alpha)}{2m}},$$

which is valid for the selected scalar threshold by the nested-class argument used in Theorem 1. For each trial, the calibration indices are sampled independently and uniformly from the target pool with replacement, with calibration size $m = 0.2T$. The confidence level is $\alpha = 0.05$ and $\varepsilon \in \{0.05, 0.10, 0.15\}$. The original PAC baseline uses the same calibration sample and the same monotone radius, but chooses its threshold directly from the calibration mistake order instead of from the fixed grid. The full sweep over $\varepsilon \in \{0.05, 0.10, 0.15\}$ is reported in Table 2 and visualized in Figure 1.

Dataset	Method	ε	Error	Violation	Budget saved
ImageNet	fixed-price monotone UCB	0.05	0.0372	0.0000	0.6296
ImageNet	original PAC	0.05	0.0377	0.0000	0.6323
ImageNet	fixed-price monotone UCB	0.10	0.0873	0.0000	0.7848
ImageNet	original PAC	0.10	0.0879	0.0000	0.7864
ImageNet	fixed-price monotone UCB	0.15	0.1372	0.0000	0.8851
ImageNet	original PAC	0.15	0.1380	0.0000	0.8866
ImageNetV2	fixed-price monotone UCB	0.05	0.0218	0.0000	0.3642
ImageNetV2	original PAC	0.05	0.0228	0.0000	0.3706
ImageNetV2	fixed-price monotone UCB	0.10	0.0718	0.0000	0.5457
ImageNetV2	original PAC	0.10	0.0726	0.0000	0.5479
ImageNetV2	fixed-price monotone UCB	0.15	0.1219	0.0000	0.6648
ImageNetV2	original PAC	0.15	0.1228	0.0000	0.6665

Table 2: Full offline sweep. Error, violation, and budget saved are averages over 500 trials. Rows are grouped by (dataset, ε) scenario and separated by light rules.

Two-source implementation. The two-source experiment constructs two complementary cheap sources from each base ImageNet table. Items are partitioned by the parity of the true class

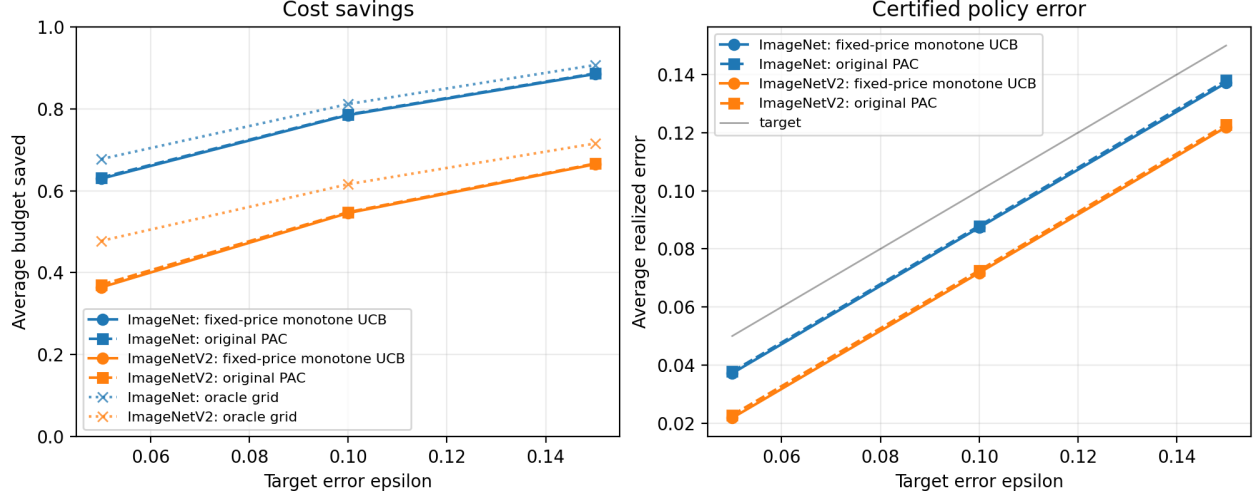


Figure 1: Simple offline sweep. The fixed-price LP rule tracks the original PAC baseline closely in both budget saved and realized error; the gray line is the target error.

label. On items inside its specialty, a source reuses the original predictor’s label and confidence; outside its specialty, it emits a random low-confidence label. The synthesized pool is fixed across trials. The calibration grid contains the expert-only policy plus 201 evenly spaced thresholds on $\min_j \hat{h}_t(j)$, the calibration size is $m = 0.2T$, and $\alpha = 0.05$. The LP method uses the monotone radius $\sqrt{\log(1/\alpha)/(2m)}$; the Bonferroni-corrected best-of-singles baseline runs both single-source baselines at $\alpha/2$ each and reports the one with larger empirical savings on the calibration sample. The full sweep over $\varepsilon \in \{0.05, 0.10, 0.15\}$ for all four methods is in Table 3 and Figure 2.

Online implementation. For each trial, the target pool is randomly permuted into a stream. The update uses $\eta = 1$, proxy floor $\rho = 0.01$, and $\lambda_1 = 1/\varepsilon$. The online routing rule uses the AI prediction when $\lambda_t \max(1 - \text{confidence}_t, \rho) < 1$ and otherwise defers to the expert. If the AI prediction is used, it is audited with probability $p \in \{0.1, 0.2, 0.5\}$. The hybrid intermittent update is $-\eta\varepsilon_0$ on expert deferrals, $\eta(\ell_t - \varepsilon_0)/p$ on audited AI decisions, and zero on unaudited AI decisions. Audited AI labels are corrected by the expert before release; unaudited AI labels are released as-is. The reported “AI error” is the counterfactual average loss before audit correction, and the reported final error is the realized loss after correction. As noted in Section 4, these runs set $\varepsilon_0 = \varepsilon$ rather than subtracting the finite-horizon slack from Corollary 2. The full sweep over $\varepsilon \in \{0.05, 0.10, 0.15\}$ and $p \in \{0.1, 0.2, 0.5\}$ is reported in Table 4 and Figure 3.

D Alternative Confidence Bounds

The Hoeffding bound (10) is simple but may be conservative. Any valid one-sided mean upper confidence bound can be substituted. For example, for a fixed λ , an empirical Bernstein bound gives

$$U_m^{\text{EB}}(\lambda; \alpha) = \hat{L}_m(\lambda) + \sqrt{\frac{2\hat{V}_m(\lambda) \log(2/\alpha)}{m}} + \frac{7 \log(2/\alpha)}{3(m-1)}, \quad (23)$$

where $\hat{V}_m(\lambda)$ is the sample variance of $H_{I_s}(\lambda)$.

When the monotone no-grid argument does not apply, the candidate prices can instead be certified

Dataset	Method	ε	Error	Violation	Budget saved
ImageNet	LP two-source monotone UCB	0.05	0.0372	0.0000	0.6296
ImageNet	Best-of-singles (Bonferroni)	0.05	0.0372	0.0000	0.3782
ImageNet	Source A monotone PAC	0.05	0.0378	0.0000	0.3743
ImageNet	Source B monotone PAC	0.05	0.0378	0.0000	0.3809
ImageNet	LP two-source monotone UCB	0.10	0.0873	0.0000	0.7848
ImageNet	Best-of-singles (Bonferroni)	0.10	0.0879	0.0000	0.4746
ImageNet	Source A monotone PAC	0.10	0.0878	0.0000	0.4715
ImageNet	Source B monotone PAC	0.10	0.0877	0.0000	0.4754
ImageNet	LP two-source monotone UCB	0.15	0.1372	0.0000	0.8851
ImageNet	Best-of-singles (Bonferroni)	0.15	0.1383	0.0000	0.5296
ImageNet	Source A monotone PAC	0.15	0.1377	0.0000	0.5273
ImageNet	Source B monotone PAC	0.15	0.1379	0.0000	0.5299
ImageNetV2	LP two-source monotone UCB	0.05	0.0218	0.0000	0.3642
ImageNetV2	Best-of-singles (Bonferroni)	0.05	0.0210	0.0000	0.2331
ImageNetV2	Source A monotone PAC	0.05	0.0227	0.0000	0.2254
ImageNetV2	Source B monotone PAC	0.05	0.0230	0.0000	0.2424
ImageNetV2	LP two-source monotone UCB	0.10	0.0718	0.0000	0.5457
ImageNetV2	Best-of-singles (Bonferroni)	0.10	0.0720	0.0000	0.3587
ImageNetV2	Source A monotone PAC	0.10	0.0727	0.0000	0.3470
ImageNetV2	Source B monotone PAC	0.10	0.0730	0.0000	0.3629
ImageNetV2	LP two-source monotone UCB	0.15	0.1219	0.0000	0.6648
ImageNetV2	Best-of-singles (Bonferroni)	0.15	0.1226	0.0000	0.4389
ImageNetV2	Source A monotone PAC	0.15	0.1234	0.0000	0.4287
ImageNetV2	Source B monotone PAC	0.15	0.1226	0.0000	0.4412

Table 3: Full two-source sweep. Error, violation, and budget saved are averages over 500 trials of calibration sampling. Rows are grouped by (dataset, ε) scenario and separated by light rules; the largest budget saved in each scenario is shown in bold.

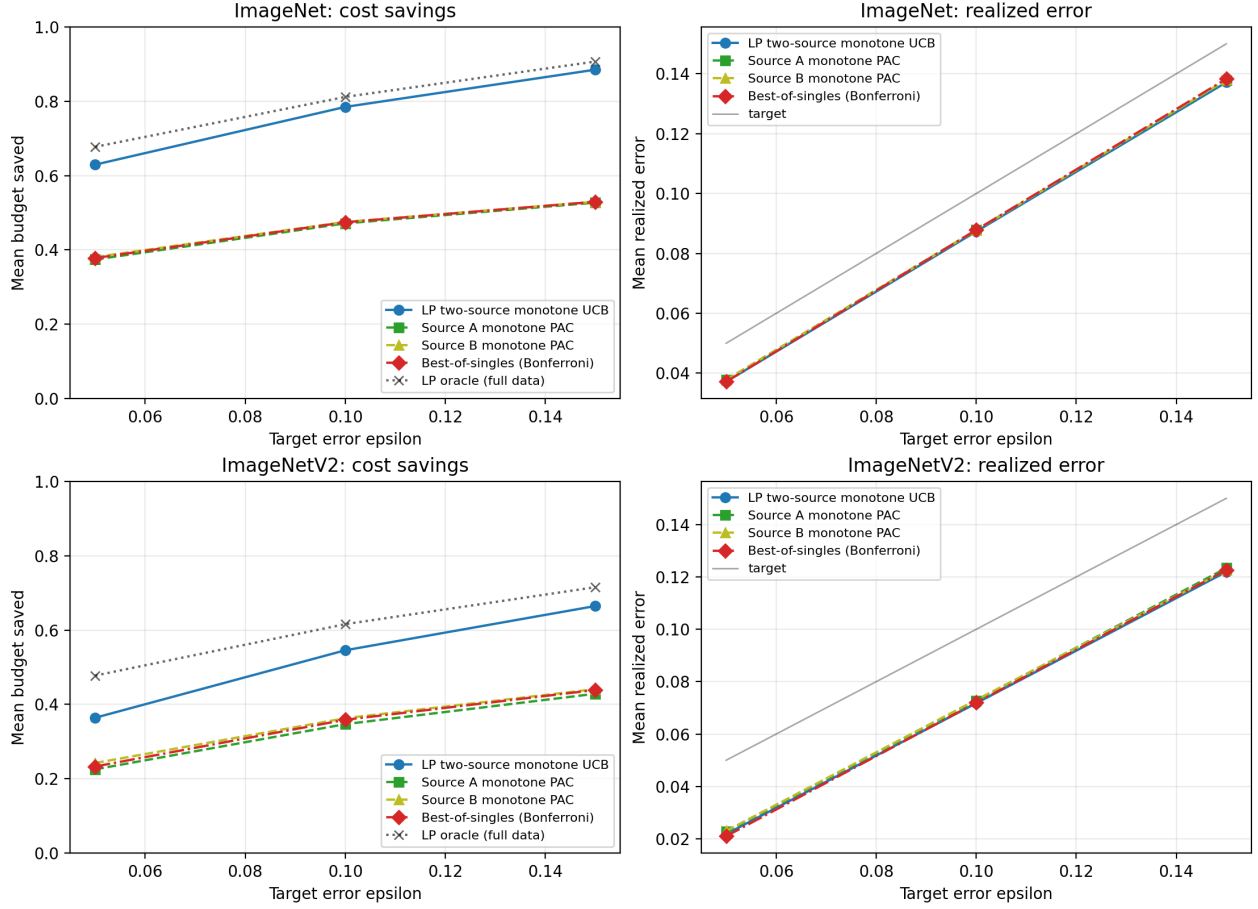


Figure 2: Two-source offline sweep. LP routing saves substantially more expert budget than the single-source baselines while keeping realized error below the target line.

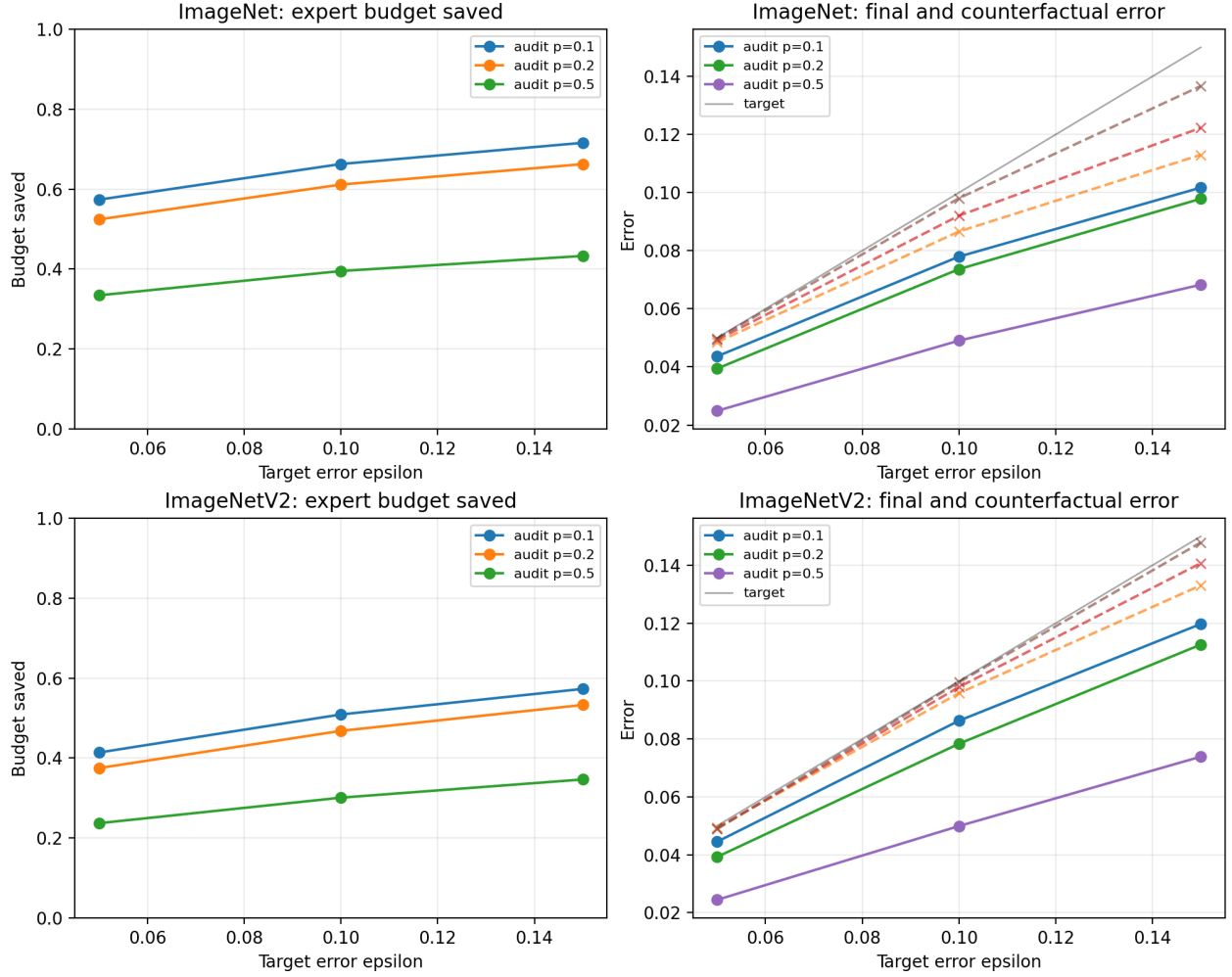


Figure 3: Online audited sweep. Smaller audit probabilities save more expert budget, while larger audit probabilities lower final released-label error by correcting more AI outputs. Dashed lines show counterfactual AI error before audit correction.

Dataset	ε	p	Final error	AI error	Violation	Budget saved
ImageNet	0.05	0.1	0.0436	0.0485	0.0100	0.5738
ImageNet	0.05	0.2	0.0394	0.0493	0.0000	0.5244
ImageNet	0.05	0.5	0.0249	0.0497	0.0000	0.3344
ImageNet	0.10	0.1	0.0780	0.0866	0.0000	0.6629
ImageNet	0.10	0.2	0.0737	0.0921	0.0000	0.6115
ImageNet	0.10	0.5	0.0490	0.0981	0.0000	0.3952
ImageNet	0.15	0.1	0.1017	0.1130	0.0000	0.7159
ImageNet	0.15	0.2	0.0979	0.1223	0.0000	0.6626
ImageNet	0.15	0.5	0.0683	0.1367	0.0000	0.4330
ImageNetV2	0.05	0.1	0.0445	0.0494	0.1840	0.4140
ImageNetV2	0.05	0.2	0.0393	0.0491	0.0100	0.3750
ImageNetV2	0.05	0.5	0.0244	0.0489	0.0000	0.2372
ImageNetV2	0.10	0.1	0.0863	0.0958	0.0400	0.5091
ImageNetV2	0.10	0.2	0.0783	0.0979	0.0000	0.4681
ImageNetV2	0.10	0.5	0.0499	0.0996	0.0000	0.3010
ImageNetV2	0.15	0.1	0.1197	0.1331	0.0000	0.5731
ImageNetV2	0.15	0.2	0.1126	0.1407	0.0000	0.5330
ImageNetV2	0.15	0.5	0.0739	0.1479	0.0000	0.3465

Table 4: Full online sweep. Final error, AI error, and budget saved are averages over 500 trials. Rows are grouped by (dataset, ε) scenario and separated by light rules.

simultaneously by a union bound. Over a finite grid Λ , the uniform Hoeffding bound

$$U_m^{\text{unif}}(\lambda; \alpha) = \hat{L}_m(\lambda) + \sqrt{\frac{\log(|\Lambda|/\alpha)}{2m}} \quad (24)$$

satisfies $\mathbb{P}\{\exists \lambda \in \Lambda : L_T(\lambda) > U_m^{\text{unif}}(\lambda; \alpha)\} \leq \alpha$.

Betting-based confidence sequences can also be used if the calibration stream is reused while expanding the candidate price set.